COMP9602: Optimization

Duality (IV)

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Next

- Duality and problem reformulation
- Partial Lagrangian relaxation
- Lagrange duality extending to problems with generalized inequality constraints

Duality and problem reformulations

- equivalent formulations of a problem can lead to very different duals
- reformulating the primal problem can be useful when the dual is difficult to derive, or uninteresting

common reformulations

- introduce new variables and equality constraints
- make explicit constraints implicit or vice-versa
- transform objective or constraint functions

  $e.g.$, replace $f_0(x)$ by $\phi(f_0(x))$ with ϕ convex, increasing

Example I

- Introducing new variables and equality constraints

minimize $f_0(Ax+b)$

- dual function is constant: $g = \inf_x L(x) = \inf_x f_0(Ax+b) = p^\star$
- we have strong duality, but dual is quite useless

reformulated problem and its dual

minimize	$f_0(y)$	maximize	$b^T\nu - f_0^*(\nu)$
subject to	$Ax+b-y=0$	subject to	$A^T\nu = 0$

Example:

minimize $\|Ax-b\|$ <=>

minimize	$\|y\|$
subject to	$y = Ax - b$

conjugate of norm on page 93, textbook

Example II

- Making explicit constraints implicit

LP with box constraints: primal and dual problem

minimize	$c^T x$	maximize	$-b^T\nu - \mathbf{1}^T\lambda_1 - \mathbf{1}^T\lambda_2$
subject to	$Ax = b$	subject to	$c + A^T\nu + \lambda_1 - \lambda_2 = 0$
	$-\mathbf{1} \preceq x \preceq \mathbf{1}$		$\lambda_1 \succeq 0, \quad \lambda_2 \succeq 0$

reformulation with box constraints made implicit

$$\begin{array}{ll} \text{minimize} & f_0(x) = \begin{cases} c^T x & -\mathbf{1} \preceq x \preceq \mathbf{1} \\ \infty & \text{otherwise} \end{cases} \\ \text{subject to} & Ax = b \end{array}$$

dual function

$$\begin{aligned} g(\nu) &= \inf_{-\mathbf{1} \preceq x \preceq \mathbf{1}} (c^T x + \nu^T (Ax - b)) \\ &= -b^T\nu - \|A^T\nu + c\|_1 \end{aligned}$$

dual problem: maximize $-b^T\nu - \|A^T\nu + c\|_1$

Partial Lagrangian relaxation

- One can take Lagrangian with respect to only a subset of the constraints
  - e.g.,

$$\begin{aligned} & \min f_0(x) \\ \text{s.t.:} \quad & f_i(x) \leq 0, i = 1, 2 \end{aligned}$$

Lagrangian	$L(x, \lambda_1, \lambda_2) = f_0(x) + \lambda_1 f_1(x) + \lambda_2 f_2(x)$
Lagrange dual	$g(\lambda_1, \lambda_2) = \min_x L(x, \lambda_1, \lambda_2)$

Or

(partial) Lagrangian	$\tilde{L}(x, \lambda) = f_0(x) + \lambda f_1(x)$
(partial) Lagrange dual	$\tilde{g}(\lambda) = \min_x \tilde{L}(x, \lambda) \text{ s.t. } f_2(x) \leq 0$

Partial Lagrangian relaxation (cont'd)

If convex problem and Slater's condition is satisfied:

$$\max_{\lambda \succeq 0} \tilde{g}(\lambda) = \min_{f_i(x) \leq 0} f_0(x)$$

let x^* be the primal optimal point, $(\lambda_1^*, \lambda_2^*)$ be dual optimal point for $g(\lambda_1, \lambda_2)$ and λ_1' be dual optimal point for $\tilde{g}(\lambda)$

$$\begin{aligned} f_0(x^*) \geq \tilde{g}(\lambda_1') \geq \tilde{g}(\lambda_1^*) &= \min_{f_2(x) \leq 0} f_0(x) + \lambda_1^* f_1(x) \\ &\geq \min f_0(x) + \lambda_1^* f_1(x) + \lambda_2^* f_2(x) \\ &= g(\lambda_1^*, \lambda_2^*) \\ &= f_0(x^*) \end{aligned}$$

The above property extensible to general case, when the partial lagrangian is derived by relaxing any subset of the constraints of a convex program

Partial Lagrangian relaxation: example

- Example:

$$\min \sum_{(i,j)\in E} c_{ij}x_{ij}$$

subject to:

$$\sum_{j:(i,j)\in E} x_{ij} - \sum_{j:(j,i)\in E} x_{ji} = \begin{cases} v, \text{for } i = s, \\ 0, \text{for all } i \in V - \{s,t\}, \\ -v, \text{for } i = t, \end{cases} \quad (1)$$

$$0 \le x_{ij} \le u_{ij}, \forall (i,j) \in E \quad (2)$$

$$\sum_{j:(i,j)\in E} x_{ij} \le O_i, \forall i \in V \qquad \lambda_i$$

$$\sum_{j:(j,i)\in E} x_{ji} \le I_i, \forall i \in V \qquad \mu_i$$

partial Lagrangian

$$\widetilde{L}(\mathrm{x},\lambda,\mu) = \sum_{(i,j)\in E} c_{ij}x_{ij} + \sum_{i\in V} \lambda_i \left(\sum_{j:(i,j)\in E} x_{ij} - O_i\right)$$

$$+ \sum_{i\in V} \mu_i \left(\sum_{j:(j,i)\in E} x_{ji} - I_i\right)$$

Partial Lagrangian relaxation: example

partial Lagrange dual:

$$\widetilde{g}(\lambda, \mu) = \min_{x} \widetilde{L}(x, \lambda, \mu)$$

s.t.: (1) (2)

dual problem:

$$\begin{array}{ll} \max & \widetilde{g}(\lambda, \mu) \\ \text{s.t.:} & \lambda \succeq 0 \\ & \mu \succeq 0 \end{array}$$

Example III

- Transform objective function

$$\text{minimize} \quad \| \, Ax - b \, \|$$

reformulation

$$\text{minimize} \ \frac{1}{2} \, \| \, y \, \|^2$$

$$\text{subject to} \quad y = Ax - b$$

dual problem on page 257, textbook

Generalized inequality

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \preceq_{K_i} 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p \end{array}$$

where $K_i \subseteq R^{k_i}$ are proper cones

Definitions for Lagrange dual are parallel to scalar case:

- Lagrange multiplier for $f_i(x) \preceq_{K_i} 0$ is vector $\lambda_i \in \mathbf{R}^{k_i}$
- Lagrangian $L : \mathbf{R}^n \times \mathbf{R}^{k_1} \times \cdots \times \mathbf{R}^{k_m} \times \mathbf{R}^p \to \mathbf{R}$, is defined as

$$L(x, \lambda_1, \cdots, \lambda_m, \nu) = f_0(x) + \sum_{i=1}^{m} \lambda_i^T f_i(x) + \sum_{i=1}^{p} \nu_i h_i(x)$$

- dual function $g : \mathbf{R}^{k_1} \times \cdots \times \mathbf{R}^{k_m} \times \mathbf{R}^p \to \mathbf{R}$, is defined as

$$g(\lambda_1, \ldots, \lambda_m, \nu) = \inf_{x \in \mathcal{D}} L(x, \lambda_1, \cdots, \lambda_m, \nu)$$

Generalized inequality

dual problem

$$\begin{array}{ll} \text{maximize} & g(\lambda_1, \ldots, \lambda_m, \nu) \\ \text{subject to} & \lambda_i \succeq_{K_i^*} 0, \quad i = 1, \ldots, m \end{array}$$

Properties:

- **lower bound property:** if $\lambda_i \succeq_{K_i^*} 0$, then $g(\lambda_1, \ldots, \lambda_m, \nu) \leq p^\star$
- weak duality: $p^\star \geq d^\star$ always
- strong duality: $p^\star = d^\star$ for convex problem with constraint qualification (for example, Slater's: primal problem is strictly feasible)
- similarly complementary slackness, KKT conditions also hold. (pp. 267, textbook)

Example

Semidefinite program

primal SDP $(F_i, G \in \mathbf{S}^k)$

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & x_1 F_1 + \cdots + x_n F_n \preceq G \end{array}$$

- Lagrange multiplier is matrix $Z \in \mathbf{S}^k$, define $< A, B >= \mathbf{tr}(AB)$
- Lagrangian $L(x, Z) = c^T x + \mathbf{tr}\left(Z(x_1 F_1 + \cdots + x_n F_n - G)\right)$
- dual function

$$g(Z) = \inf_x L(x, Z) = \begin{cases} -\mathbf{tr}(GZ) & \mathbf{tr}(F_i Z) + c_i = 0, \quad i = 1, \ldots, n \\ -\infty & \text{otherwise} \end{cases}$$

dual problem

$$\begin{array}{ll} \text{maximize} & -\mathbf{tr}(GZ) \\ \text{subject to} & Z \succeq 0, \quad \mathbf{tr}(F_i Z) + c_i = 0, \quad i = 1, \ldots, n \end{array}$$

$p^\star = d^\star$ if primal SDP is strictly feasible ($\exists x$ with $x_1 F_1 + \cdots + x_n F_n \prec G$)

- Reference
  - Chapter 5.7, 5.9, Convex Optimization.
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